

Final Year Project Long Proposal

RetainIQ: Intelligent Customer Retention & Analytics System

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Revision History

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Muhammad Fahad Akram	Nisar Azim	14-Sep-2026	Section 1	1.0
Muhammad Shoaib Ali	Nisar Azim	17-Sep-2026	Section 2	1.1
Muhammad Mahad Atique	Nisar Azim	20-Sep-2026	Section 3	1.2
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1. Project Stream

- [x] Web-Based FYP
- [ ] Desktop Application
- [ ] Mobile App
- [ ] Game
- [ ] Hardware-Based

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2. Introduction

Customer retention is a critical concern for modern businesses because long-term profitability depends not only on acquiring new customers but also on understanding and preserving existing customer relationships. Organizations generate large volumes of customer and transaction data through purchases, interactions, and service activities, yet this information is often fragmented or underutilized. Traditional reporting tools mainly describe past performance and do not always identify which customers are becoming inactive or why their behavior is changing. Recent customer platforms therefore increasingly combine unified customer profiles, segmentation, predictive analytics, and automated engagement to support data-driven retention decisions [1, 2]. RetainIQ is proposed as an intelligent web-based analytics system that converts historical customer and transaction data into actionable retention intelligence.

The system will provide an end-to-end analytical workflow covering customer and transaction data management, preprocessing and validation, purchase analytics, RFM analysis, customer segmentation, churn prediction, and retention risk scoring. RetainIQ will further provide explainable customer risk analysis so that users can understand the major factors contributing to a customer's predicted risk instead of relying on an unexplained model output. A Customer 360 profile will consolidate behavioral indicators, purchase history, segment information, and risk results in one view. A What-If Risk Simulator will allow users to change selected behavioral values and observe how the estimated risk may respond, while the recommendation and intervention module will suggest suitable retention actions. Finally, dashboards and reports will summarize customer health, segment trends, churn risk, and retention outcomes for decision-makers.

3. Project Description

RetainIQ: Intelligent Customer Retention & Analytics System is a web-based decision-support platform designed to help businesses analyze customer purchasing behavior and identify customers who may be at risk of becoming inactive or churning. The project

will accept structured customer and transaction records, validate and preprocess them, and transform them into analytical features suitable for descriptive and predictive analysis. Purchase analytics and RFM analysis will measure recency, frequency, and monetary behavior, while customer segmentation will group customers with similar patterns. Machine-learning-based churn prediction and a retention risk score will then help prioritize customers who require attention. The objective is to make retention analysis accessible through a single integrated system rather than requiring separate spreadsheets, scripts, dashboards, and CRM tools.

A major focus of RetainIQ is interpretability and actionability. For each customer, the system will present a Customer 360 profile containing purchase indicators, RFM values, segment membership, churn probability, risk category, and explanations of influential risk factors. The What-If Risk Simulator will support scenario analysis by allowing authorized users to adjust selected customer-behavior variables and compare the resulting predicted risk. Based on the customer's segment, behavior, and risk level, the system will provide retention recommendations and intervention suggestions that can guide business follow-up. Management dashboards will aggregate results across the customer base and present trends, segment distributions, high-risk customers, and retention-oriented metrics. In this way, RetainIQ combines analytics, prediction, explanation, simulation, and intervention support in one coherent workflow.

4. Real World Problem

Businesses frequently possess years of customer and transaction data but still struggle to recognize declining engagement before a customer stops purchasing. Customer information may be distributed across sales systems, spreadsheets, and reporting tools, making it difficult to obtain a consistent view of individual behavior. Basic dashboards can show revenue or transaction totals, but they may not reveal which customers are becoming less frequent, how recently they purchased, or which behavioral patterns indicate churn risk. Retention teams therefore risk reacting only after inactivity has already occurred. Existing enterprise platforms offer useful capabilities such as unified profiles, segmentation, health scoring, and churn prediction [1, 2, 5, 4], but these capabilities can be distributed across larger CRM or marketing ecosystems and may not provide a compact academic workflow combining RFM analysis, explainable churn risk, interactive what-if simulation, and retention recommendations. The real-world need is therefore for an integrated system that cleans customer data, analyzes purchase behavior, predicts retention risk, explains the prediction, and translates the result into understandable interventions. RetainIQ addresses this need by turning historical transaction records into customer-level intelligence that supports earlier, evidence-based retention decisions.

5. Problem Statement

Organizations often lack an integrated and interpretable mechanism for converting raw customer transaction data into timely retention decisions. Existing data may reveal purchase history, but without systematic preprocessing, RFM analysis, segmentation, churn prediction, risk scoring, and explanation, decision-makers cannot easily identify which customers require intervention or understand the reasons behind their risk. This creates a gap between descriptive customer reporting and proactive retention management. RetainIQ aims to address this gap by providing a unified analytics system that predicts customer churn risk, explains influential factors, supports what-if analysis, and recommends retention interventions through customer-level profiles and management dashboards.

6. Application Review

This section reviews five existing customer analytics and retention platforms. The review focuses on capabilities relevant to RetainIQ, including unified customer data, segmentation, churn or health-risk analysis, explainability, and retention action support.

6.1 Salesforce Data 360 [1]

Salesforce Data 360 unifies structured and unstructured enterprise data, supports identity resolution, customer profiles, segmentation, calculated insights, predictive models, and activation workflows [1].

- **Key characteristic 1:** Unified customer profiles and identity resolution across multiple data sources.
- **Key characteristic 2:** Dynamic segmentation and predictive insights for customer activation.

The limitations of the application are listed below.

- **Limitation 1:** It is a broad enterprise data platform rather than a focused RFM-to-retention academic workflow.
- **Limitation 2:** RetainIQ's dedicated what-if churn-risk simulator is not presented as a core equivalent feature in the reviewed Data 360 material.

6.2 Microsoft Dynamics 365 Customer Insights [2]

Dynamics 365 Customer Insights supports customer segmentation and predictive models, including transactional and subscription churn models. Churn outputs include a probability score and influential factors that can be used to build high-risk segments [2].

- **Key characteristic 1:** Transactional churn prediction and customer risk scoring.
- **Key characteristic 2:** Segmentation based on behavioral, transactional, and pre-

dictive attributes.

The limitations of the application are listed below.

- **Limitation 1:** It is part of a larger enterprise ecosystem with broader configuration requirements.
- **Limitation 2:** The reviewed documentation does not describe a dedicated interactive what-if risk simulator equivalent to RetainIQ.

6.3 Optimove [3]

Optimove is a customer-led marketing and retention platform that uses dynamic micro-segmentation, predictive behavior modeling, churn analysis, and action optimization to support proactive retention [3].

- **Key characteristic 1:** Predictive segmentation and churn-oriented behavioral modeling.
- **Key characteristic 2:** Retention action optimization based on customer and segment behavior.

The limitations of the application are listed below.

- **Limitation 1:** The platform emphasizes marketing orchestration rather than a transparent educational pipeline from preprocessing through RFM and churn modeling.
- **Limitation 2:** A user-controlled what-if churn-risk simulator is not identified in the reviewed material.

6.4 Zoho CRM with Zia [4]

Zoho CRM's Zia can predict customer churn, display a numerical churn probability, and present factors that induce or prevent churn. The prediction can also be used in reports, filters, and automated workflows [4].

- **Key characteristic 1:** Churn probability scoring with reasons contributing to the prediction.
- **Key characteristic 2:** Workflow actions and reports can use churn prediction results.

The limitations of the application are listed below.

- **Limitation 1:** Churn prediction availability depends on specified Zoho CRM editions and organizational requirements.
- **Limitation 2:** The reviewed feature is CRM-centric and does not expose RetainIQ's full RFM, segmentation, what-if, and intervention workflow as one dedicated module chain.

6.5 HubSpot Customer Success Workspace [5]

HubSpot provides customer health scores, customer-success views, product-usage insights, automated alerts, and workflows to help teams identify at-risk accounts and prioritize proactive outreach [5].

- **Key characteristic 1:** Configurable health scores based on customer attributes and behavioral events.
- **Key characteristic 2:** Customer-level context, health trends, and proactive outreach support.

The limitations of the application are listed below.

- **Limitation 1:** Health scoring is primarily configured within a CRM/customer-success workspace rather than as a standalone transaction-analytics pipeline.
- **Limitation 2:** A dedicated what-if simulator for churn probability is not described in the reviewed customer-success material.

6.6 RetainIQ (Proposed System)

RetainIQ combines transaction preprocessing, purchase analytics, RFM analysis, segmentation, churn prediction, retention risk scoring, explainable risk analysis, Customer 360 profiles, what-if simulation, recommendations, and dashboards in a single web-based system.

- **Key characteristic 1:** End-to-end customer-retention analytics from raw transaction data to explainable risk and intervention.
- **Key characteristic 2:** Interactive what-if risk simulation and customer-level retention recommendations.

Table 1: Applications Comparison

Feature	Salesforce [1]	Dynamics 365 [2]	Optimove [3]	Zoho Zia [4]	HubSpot [5]	Proposed System
Customer/Transaction Data Management	✓	✓	✓	✓	✓	✓
Data Preprocessing/Validation	✓	✓	✓	✗	✗	✓
RFM Analysis	✗	✗	✗	✗	✗	✓
Customer Segmentation	✓	✓	✓	✓	✓	✓
Churn Prediction	✓	✓	✓	✓	✗	✓
Retention/Health Risk Scoring	✓	✓	✓	✓	✓	✓
Explainable Risk Factors	✗	✓	✗	✓	✓	✓
Customer 360 Profile	✓	✓	✓	✓	✓	✓
What-If Risk Simulator	✗	✗	✗	✗	✗	✓
Retention Recommendations/Intervention	✓	✓	✓	✓	✓	✓
Analytics Dashboard/Reporting	✓	✓	✓	✓	✓	✓

7. Scope

The scope of RetainIQ covers a web-based customer-retention analytics system that processes structured customer and transaction data and converts it into descriptive, predictive, and actionable insights. The system is intended for business analysts, managers, and authorized staff who need to understand purchasing behavior, identify at-risk customers, and plan retention interventions.

7.1 Included Aspects

- Import, storage, management, validation, and preprocessing of customer and transaction datasets.
- Purchase analytics, RFM analysis, customer segmentation, churn prediction, and retention risk scoring.
- Explainable risk analysis and a consolidated Customer 360 profile.
- What-if risk simulation using selected customer behavioral variables.
- Retention recommendations, intervention tracking, dashboards, and reporting.

7.2 Excluded Aspects

- Automatic execution of paid advertising or external marketing campaigns.
- Real-time integration with every third-party CRM, ERP, or e-commerce platform.
- Financial credit scoring or medical/clinical prediction.
- Guaranteed prevention of churn; the system provides decision support based on available data.

7.3 Modules Description

Module 1: Customer & Transaction Data Management

Centralized CRUD operations and structured storage for customer and transaction records.

Module 2: Data Preprocessing & Validation

Missing-value handling, duplicate detection, type/range validation, and preparation of analysis-ready data.

Module 3: Customer Purchase Analytics

KPIs and trends such as purchase frequency, order value, revenue contribution, and purchase history.

Module 4: RFM Analysis

Calculation and scoring of Recency, Frequency, and Monetary behavior for each customer.

Module 5: Customer Segmentation

Grouping customers into meaningful behavioral clusters or business segments.

Module 6: Customer Churn Prediction

Machine-learning prediction of the likelihood that a customer will churn or become inactive.

Module 7: Retention Risk Scoring

Conversion of model outputs and customer indicators into understandable retention-risk levels.

Module 8: Explainable Customer Risk Analysis

Presentation of influential factors that increase or decrease an individual customer's predicted risk.

Module 9: Customer 360° Profile

Unified profile combining identity, transactions, RFM, segment, risk, explanations, and intervention history.

Module 10: What-If Risk Simulator

Scenario analysis that estimates how selected behavioral changes may affect predicted retention risk.

Module 11: Retention Recommendation & Intervention

Generation of context-aware retention suggestions and recording of planned/completed interventions.

Module 12: Retention Analytics Dashboard & Reporting

Management dashboards and exportable summaries for churn, risk, segments, purchases, and retention performance.

8. Modules

Table 2 provides a breakdown of the project modules and assigned students based on deadlines, showcasing their roles and responsibilities.

Table 2: RetainIQ Module Distribution and Task Allocation

Deadline	**Modules**	**Student**
P1 Mid	1: Customer & Transaction Data Management	Muhammad Fahad Akram (23108173)
	2: Data Preprocessing & Validation	Muhammad Shoaib Ali (23108175)
	3: Customer Purchase Analytics	Muhammad Mahad Atique (23108257)
P1 Final	4: RFM Analysis	Muhammad Fahad Akram (23108173)
	5: Customer Segmentation	Muhammad Shoaib Ali (23108175)
	6: Customer Churn Prediction	Muhammad Mahad Atique (23108257)
P2 Mid	7: Retention Risk Scoring	Muhammad Fahad Akram (23108173)
	8: Explainable Customer Risk Analysis	Muhammad Shoaib Ali (23108175)
	9: Customer 360° Profile	Muhammad Mahad Atique (23108257)
P2 Final	10: What-If Risk Simulator	Muhammad Fahad Akram (23108173)
	11: Retention Recommendation & Intervention	Muhammad Shoaib Ali (23108175)
	12: Retention Analytics Dashboard & Reporting	Muhammad Mahad Atique (23108257)

9. Development Environment

Tools: Visual Studio Code/PyCharm, Jupyter Notebook, Git and GitHub, Postman.
Platform: Web-based application using Python for analytics and machine learning with a modern browser-based front end.
DBMS: PostgreSQL or MySQL for structured customer, transaction, model-output, and intervention data.
Libraries/Frameworks: Pandas, NumPy, scikit-learn, SHAP, Flask/FastAPI or Django, and a JavaScript-based charting library for dashboards.

10. Significance

RetainIQ is significant because it connects customer analytics with practical retention decision support. Instead of limiting analysis to historical sales summaries, the system will help users identify customers whose purchasing behavior indicates declining engagement, quantify their risk, and understand the factors influencing that risk. This can support earlier and more consistent intervention while also making analytical results easier for non-technical decision-makers to interpret.

- **Integrated Analytics:** Combines purchase analytics, RFM, segmentation, churn prediction, and reporting in one system.
- **Explainability:** Provides customer-level explanations for risk predictions rather than only a probability score.
- **Decision Support:** Links identified risk to recommendations and intervention records.
- **Scenario Analysis:** The what-if simulator helps users explore how hypothetical behavioral changes may influence risk.
- **Customer-Centric View:** A Customer 360 profile consolidates the most relevant information for each customer.

11. Tools and Techniques

RetainIQ will use a layered web architecture in which customer and transaction data are stored in a relational database, processed by a Python analytics layer, and presented through a browser-based user interface. Data preprocessing will use Pandas and NumPy for cleaning, transformation, validation, and feature engineering. RFM analysis will derive recency, frequency, and monetary indicators from transaction history, while segmentation can use rule-based RFM groups and/or clustering techniques such as K-Means where appropriate.

For churn prediction, supervised machine-learning algorithms such as Logistic Regression, Random Forest, Gradient Boosting, or XGBoost-style models may be evaluated using suitable classification metrics. The final model will be selected based on validation performance and interpretability rather than accuracy alone. Retention risk scoring will translate predicted probabilities into business-friendly risk categories. Explainable AI techniques such as SHAP may be used to identify the contribution of individual features to customer-level predictions. The What-If Risk Simulator will reuse the trained prediction pipeline to estimate risk under user-specified changes while clearly presenting the result as a scenario rather than a guaranteed outcome.

The application layer can be implemented using Flask, FastAPI, or Django with

RESTful services, while the front end will provide forms, tables, Customer 360 views, and interactive dashboards. PostgreSQL or MySQL will maintain normalized records for customers, transactions, model results, recommendations, and interventions. GitHub will support collaborative version control, and Postman will be used for API testing. Unit, integration, validation, and model-evaluation tests will be performed to ensure that data processing and analytical outputs remain consistent across the twelve modules.

References

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[2] Microsoft. (2026, Jul. 13). "Predict transaction churn – Dynamics 365 Customer Insights." [Online]. Available: https://learn.microsoft.com/en-us/dynamics365/customer-insights/data/predict-transactional-churn. [Accessed: Oct. 7, 2026].

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